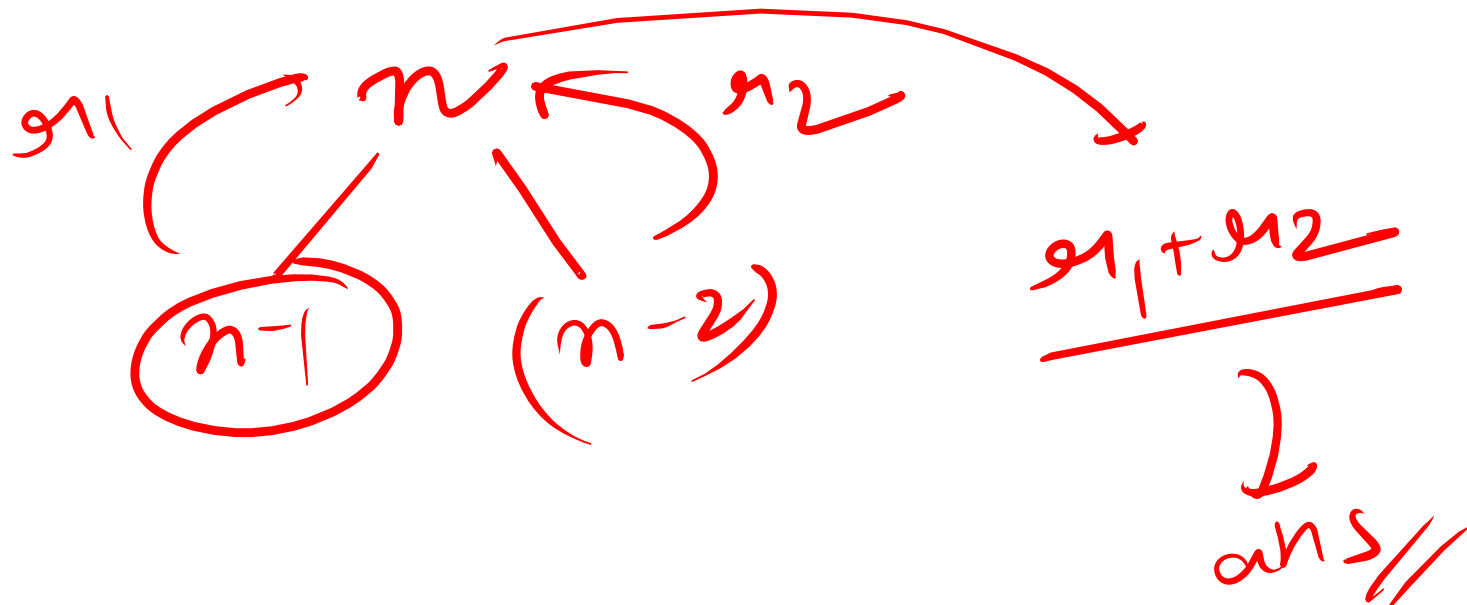


✓ ✓ Fibonacci series

<u>0</u>	<u>1</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>5</u>	<u>...</u>
0	1	2	3	4	5	6
						∞

→

$$\underline{f(n)} = \underline{f(n-1)} + \underline{f(n-2)}$$



2
c

fib(n) {

if (n == 0 || n == 1)
return n;

u1 = fib(n-1);

u2 = fib(n-2);

return u1 + u2;

}

main → fib(4) → n = 4

fib(3) → n = 3

fib(2) → n = 2

fib(1) → n = 1

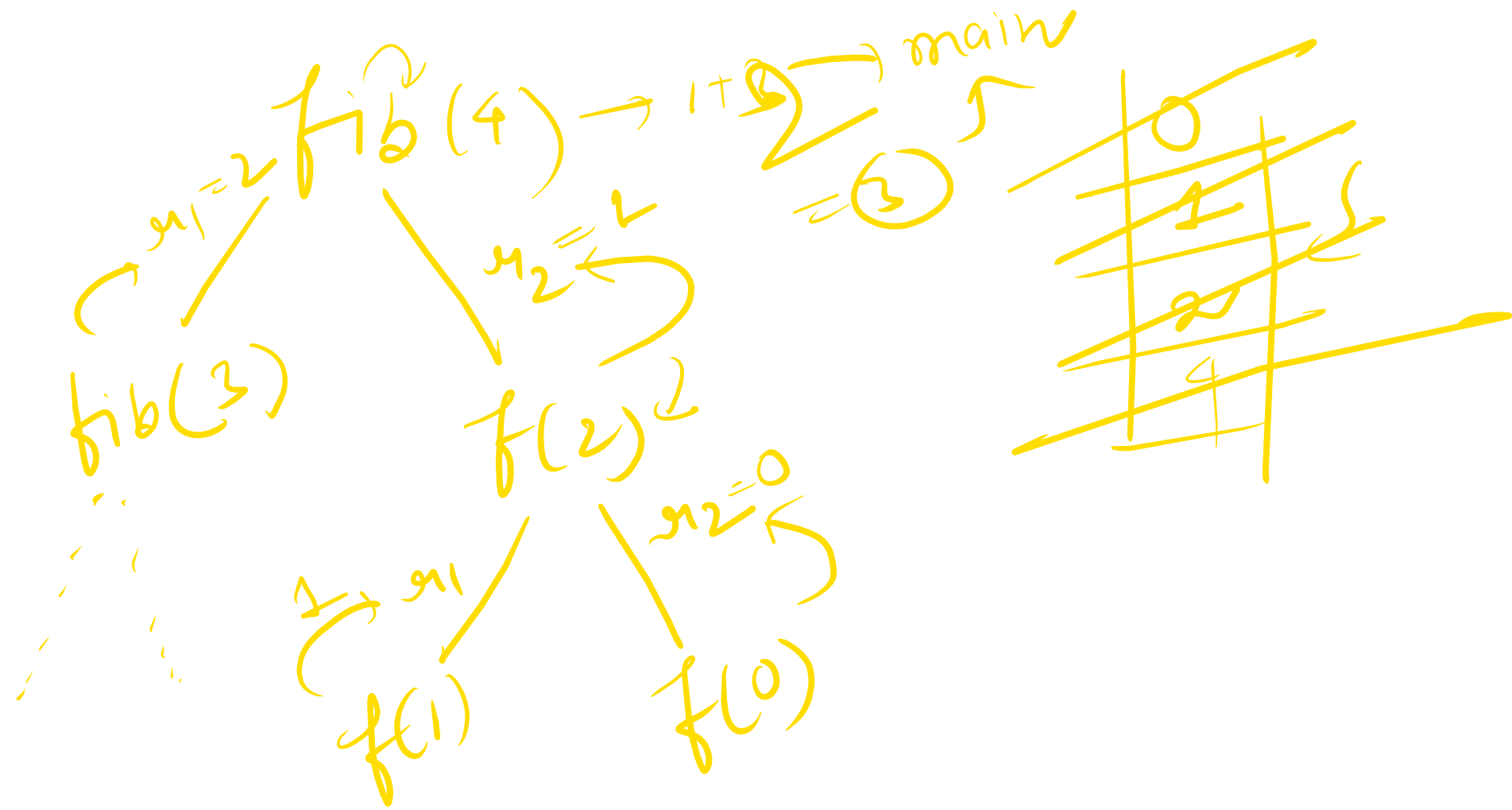
fib(0)
n = 0

fib(1)
n = 1

2^n
x
n

0	1
1	2
2	3
3	4
4	5

O(n)
→ SC



$$\frac{2^1}{2^0} = 2 / \frac{2^2}{2^1} = 2 \dots$$

$$0 \dots n-1$$

$$2^0 + 2^1 + 2^2 + 2^3 \dots + 2^{n-1}$$

GP $\rightarrow$ $n=2$

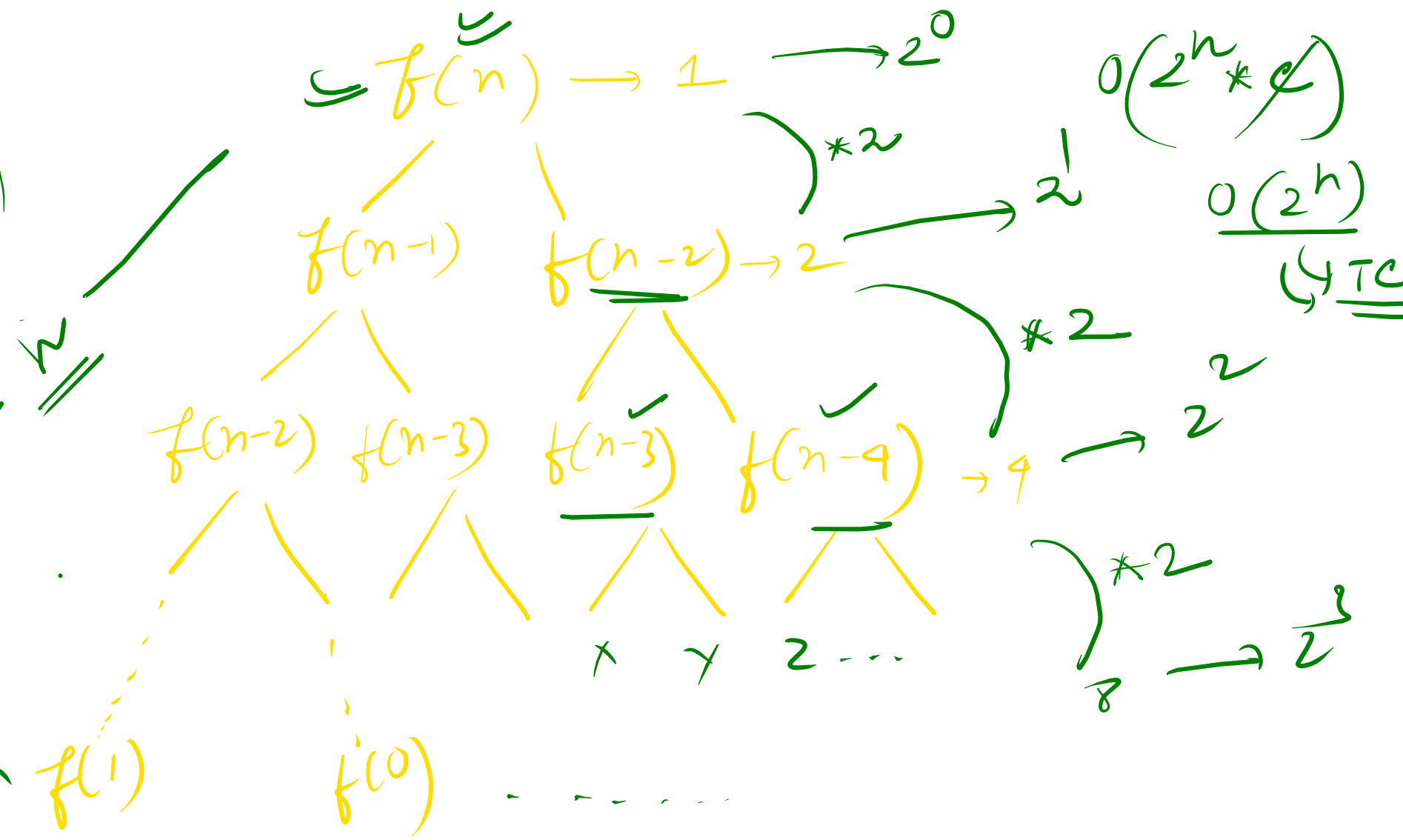
$$\frac{a \cdot (n^n - 1)}{n - 1}$$

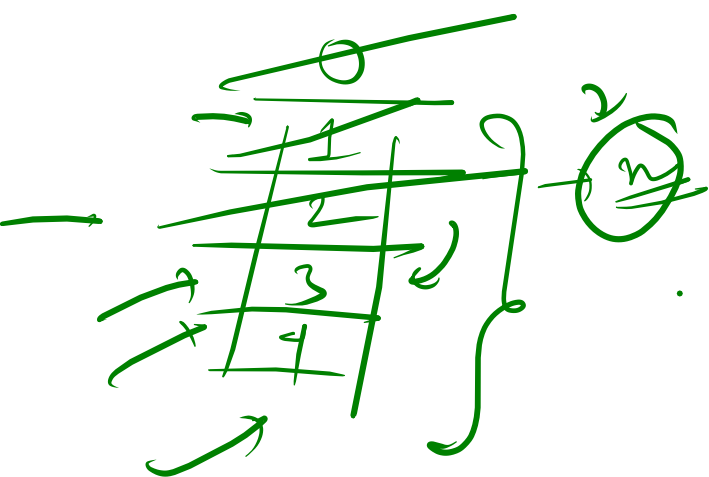
$$\frac{1 \cdot (2^n - 1)}{2 - 1}$$

$\downarrow$

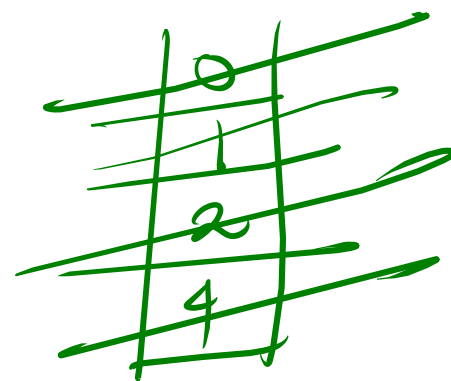
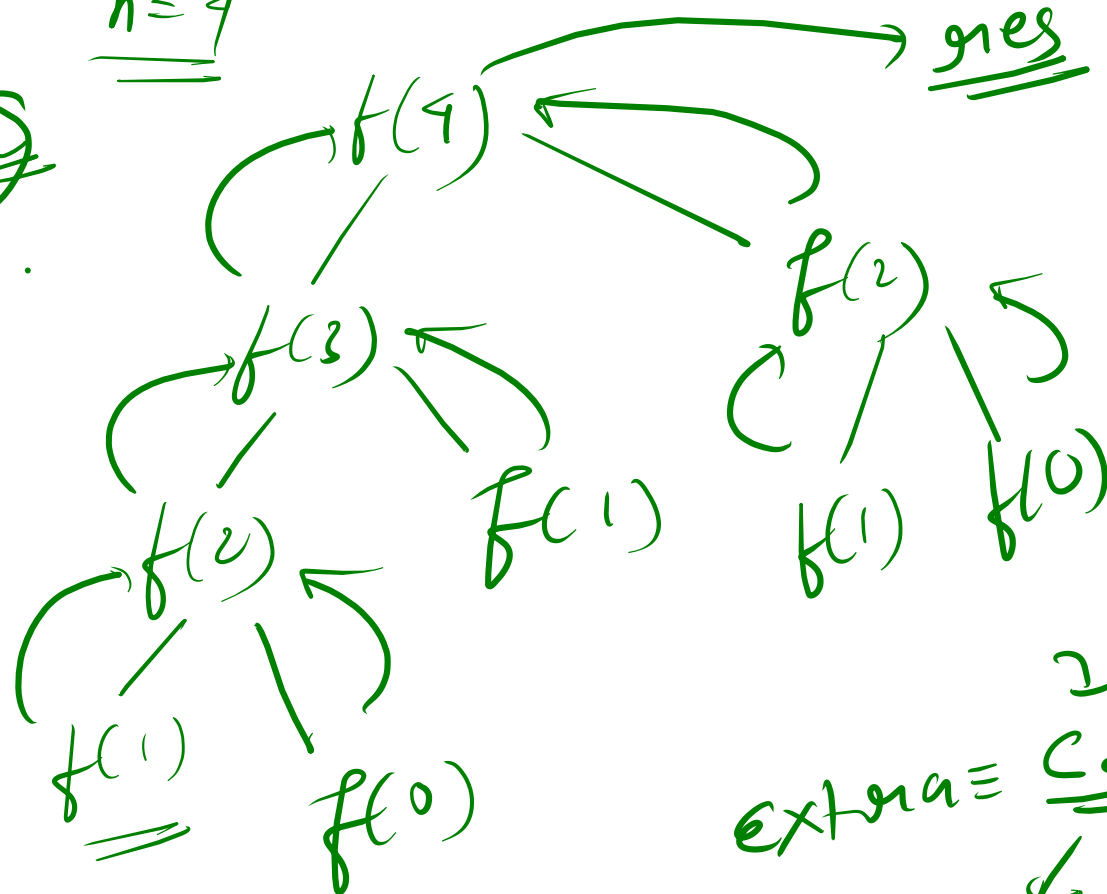
$$\underline{\underline{2^n - 1}}$$

$$\underline{\underline{2^{n-1}}}$$





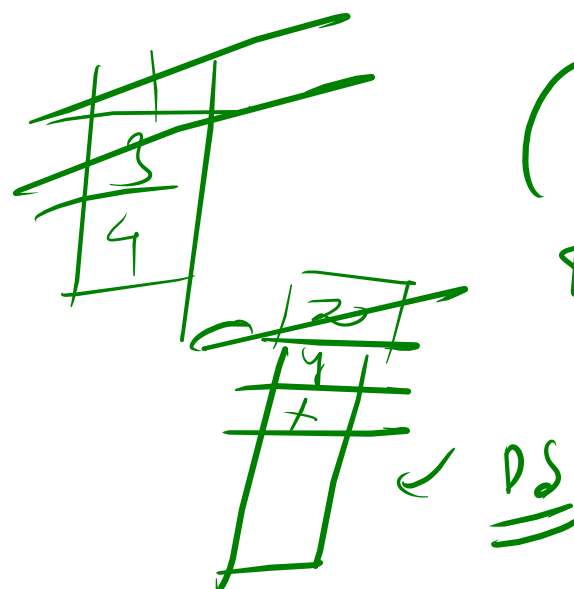
$n=9$

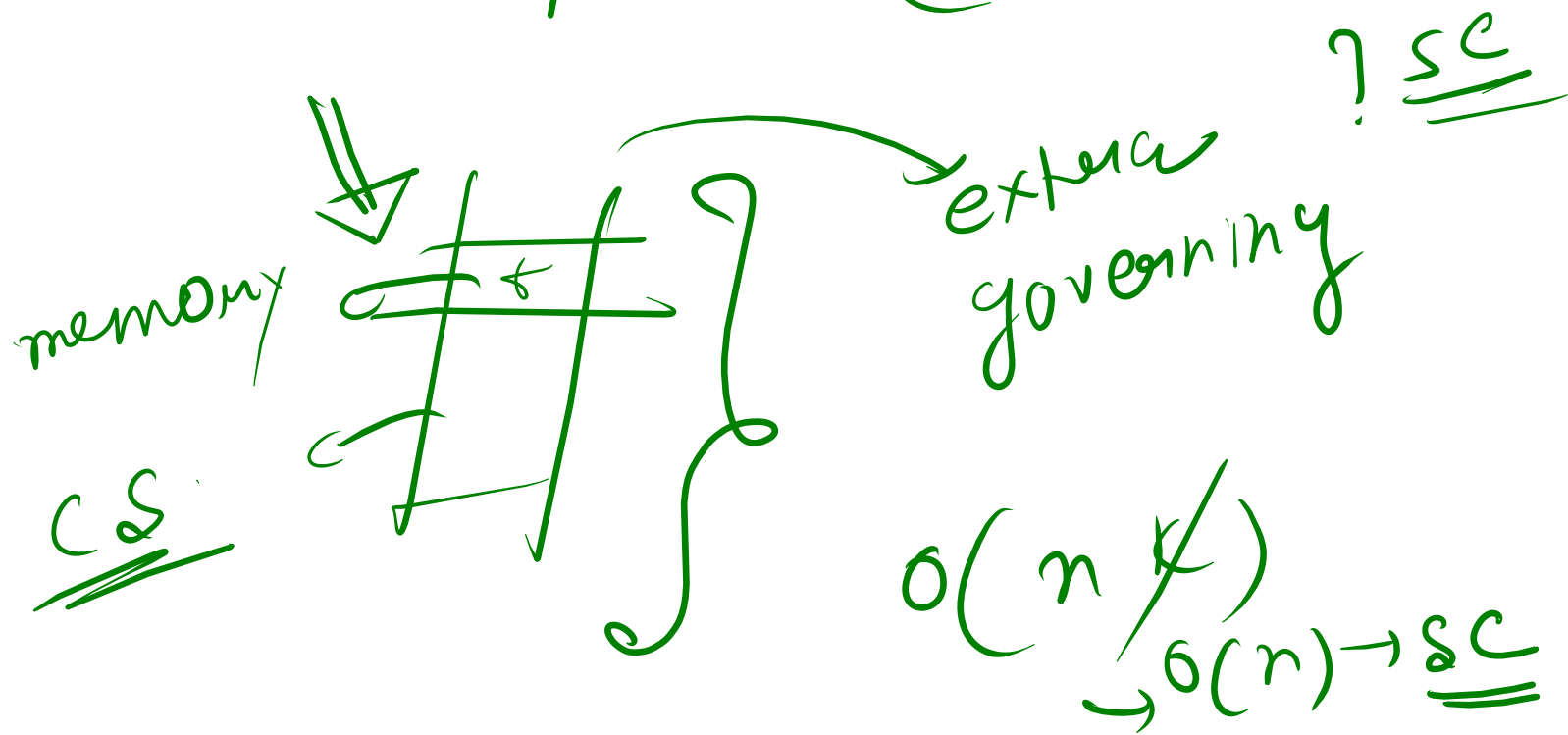
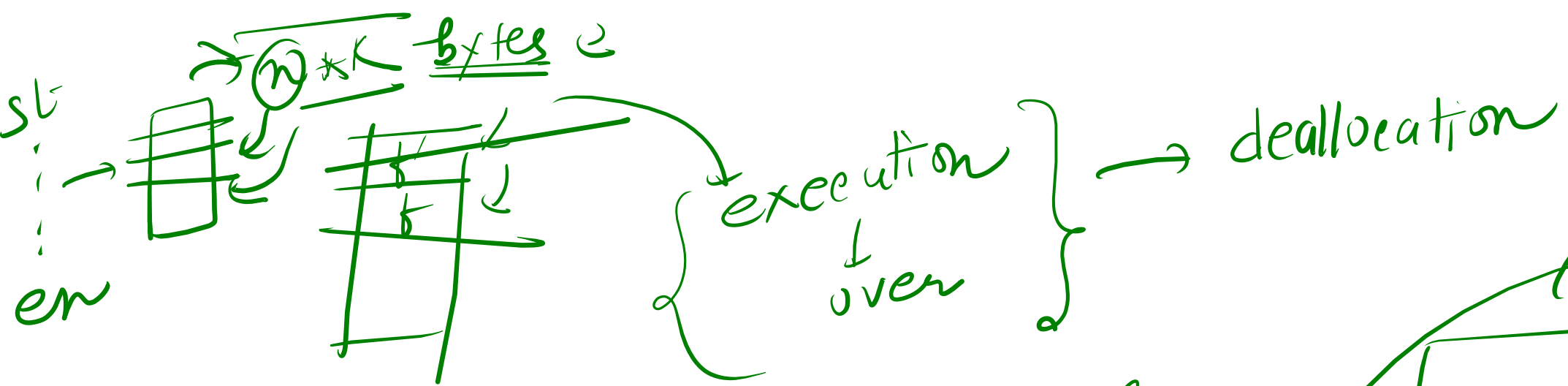


$\sim k$ bytes

extra = $\frac{C \cdot S}{n}$

$d(n*/k) \frac{d(n)}{SC}$





$$x=1$$

$$y=1$$

0 1 1 2 3 5 8 13 21 34 ...

nth even fibonacci { 0 2 8 34 ... }

$F(n) \rightarrow$ nth even fibonacci

$$\underline{\underline{F(n)}} = \underline{x F(n-1)} + \underline{F(n-2) y} ?$$

even $f(\overline{n}) = \vec{x} \cdot \underline{f(n-3)} + \vec{y} \cdot \underline{f(n-6)}$?

derive:

$$f(n) = f(n-1) + f(n-2)$$

$$= 2f(n-2) + f(n-3) \leftarrow$$

$$= 2 * (f(n-3) + f(n-4)) + f(n-3)$$

$$= 2f(n-3) + 2f(n-4) + f(n-3)$$

$$N = 4 * (N-1) + 1 * (N-2)$$

$$f(\underline{N}) =$$

$$3f(n-3) + 2f(\underline{n-4}) + f(n-5)$$

$$3f(n-3) + \boxed{f(n-4) + f(n-5)}$$

$$4f(\underline{N-1}) + f(\underline{n-2}) + f(n-6)$$

$$4f(\underline{n-3}) + f(n-6)$$

$$x=4$$

$$y=1$$

$$z=1$$

$N \rightarrow \begin{matrix} \text{even} \\ \text{fib} \end{matrix}$

$$f(N) = 4f(N-1) + f(N-2)$$

Original fib
series.

Handwritten diagram illustrating a sequence of terms and their corresponding exponents. The top part shows a sequence of 'e' characters, with curly braces indicating groups. The bottom part shows a sequence of terms: x^{n-6} , x^{n-3} , and x^{n+1} , with a large bracket underneath. An arrow points from the label 'nth' to the x^{n+1} term.

0 2 8 39 $\frac{a}{??}$

$$8 = 2 \times 4 + 0 \quad \checkmark$$

$$34 = 8 \times 4 + 2 \quad \checkmark$$

$$a_2 = 4 * 34 + 8 = 144 \dots$$

$n \rightarrow$

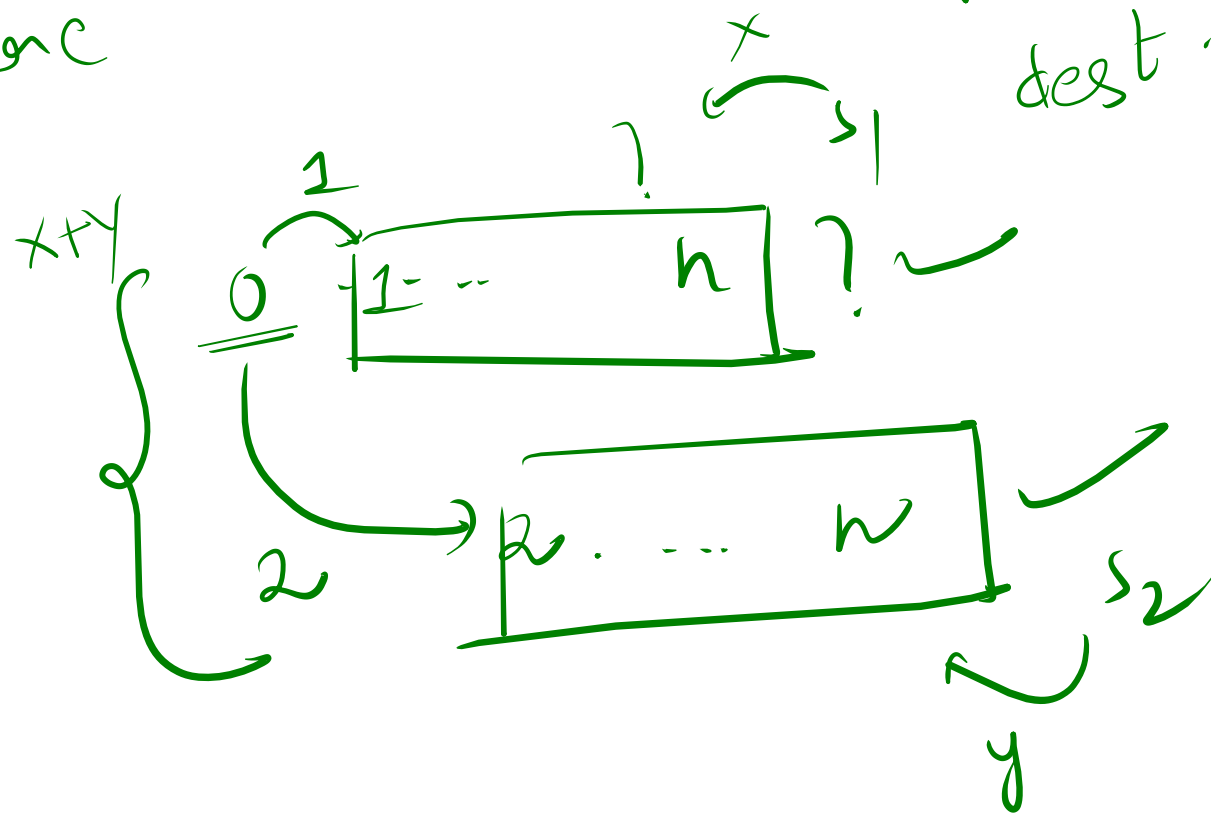
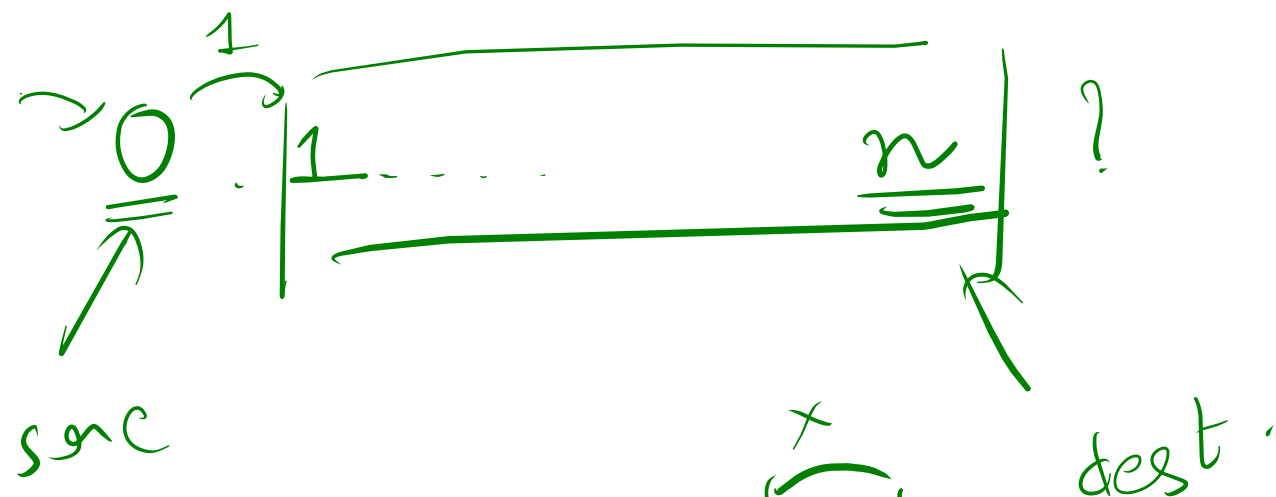
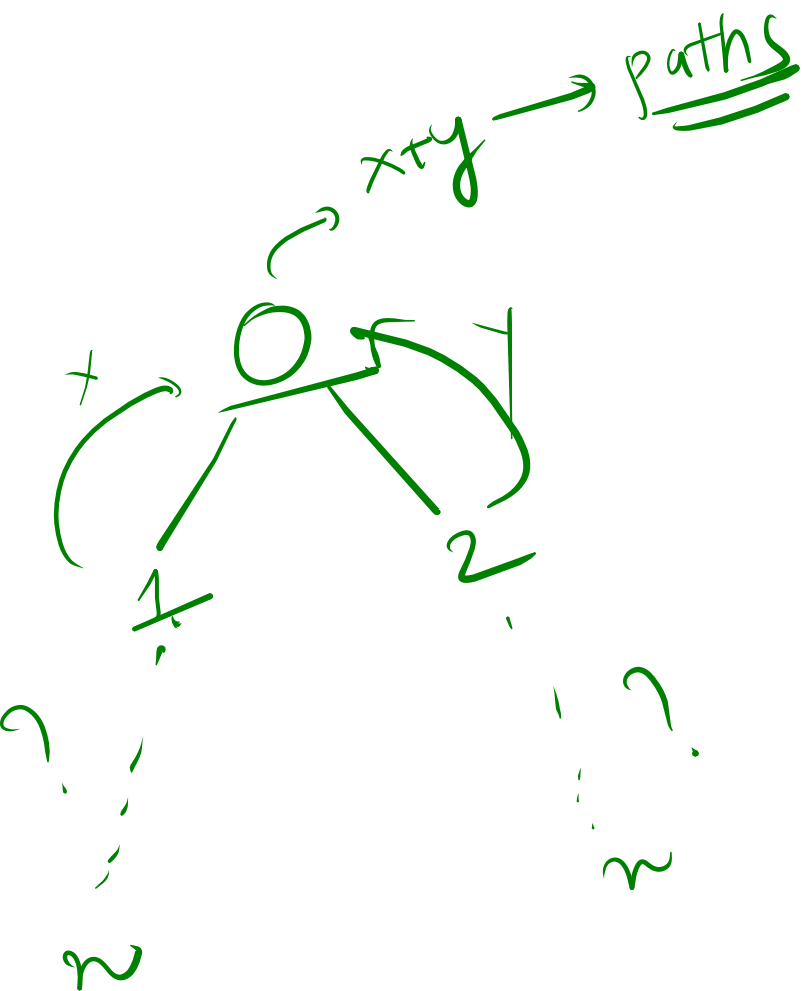
$$f(n) = 4 * f(n-3) + f(n-6)$$

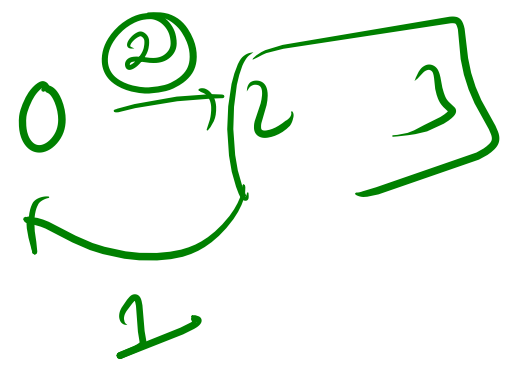
$\downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow$
 $n \qquad \qquad \qquad n-1 \qquad \qquad \qquad n-2$

Even

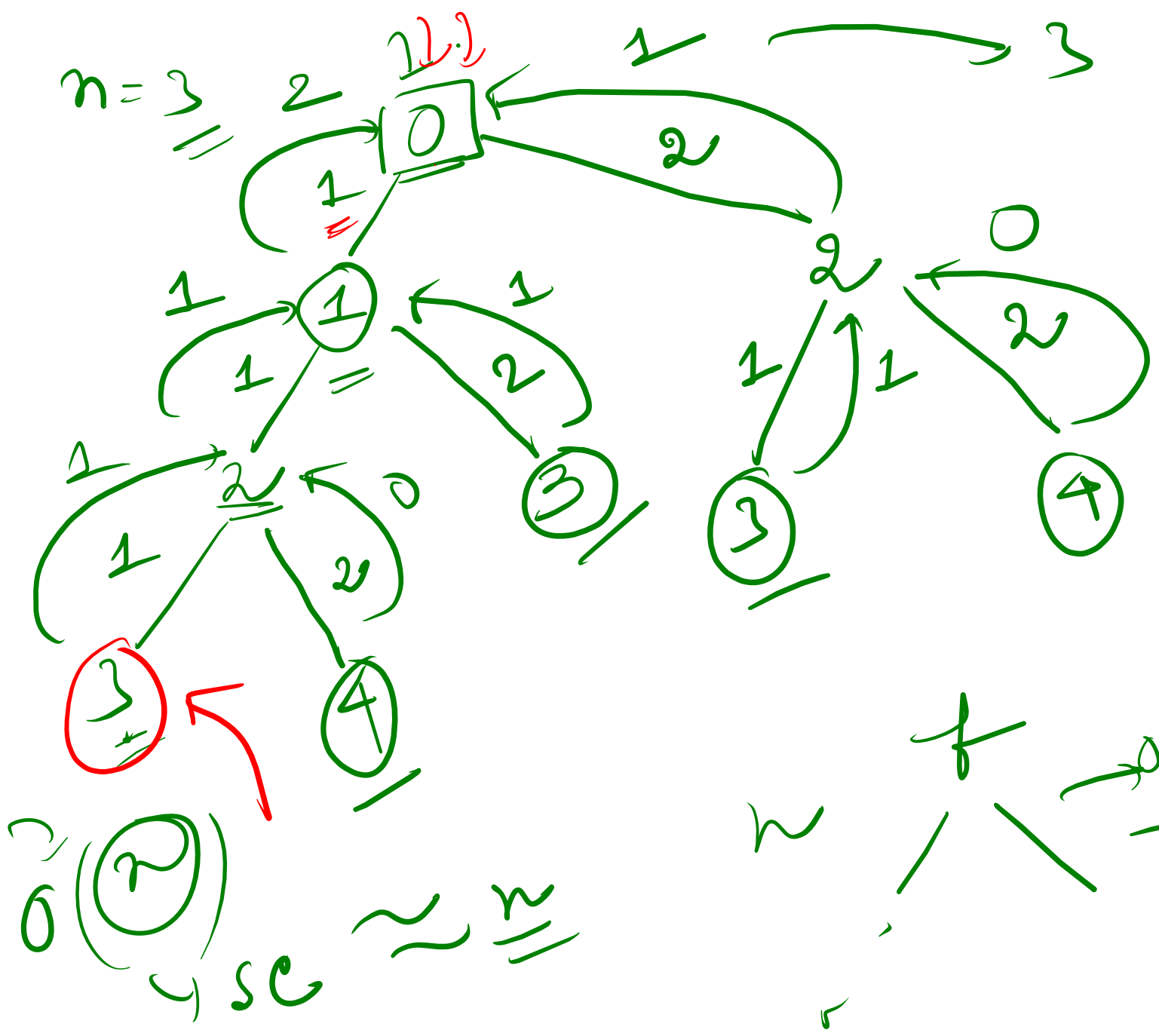
$$F(N) = 4 * F(N-1) + F(N-2)$$

clear!

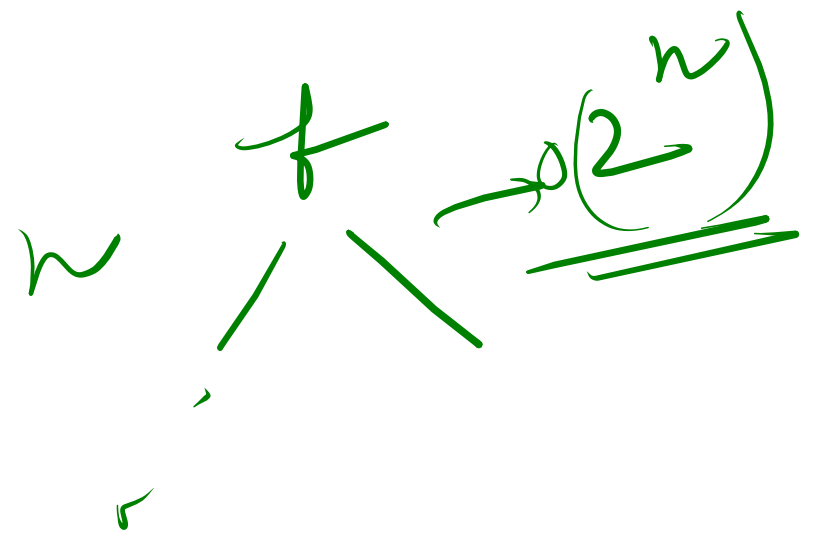


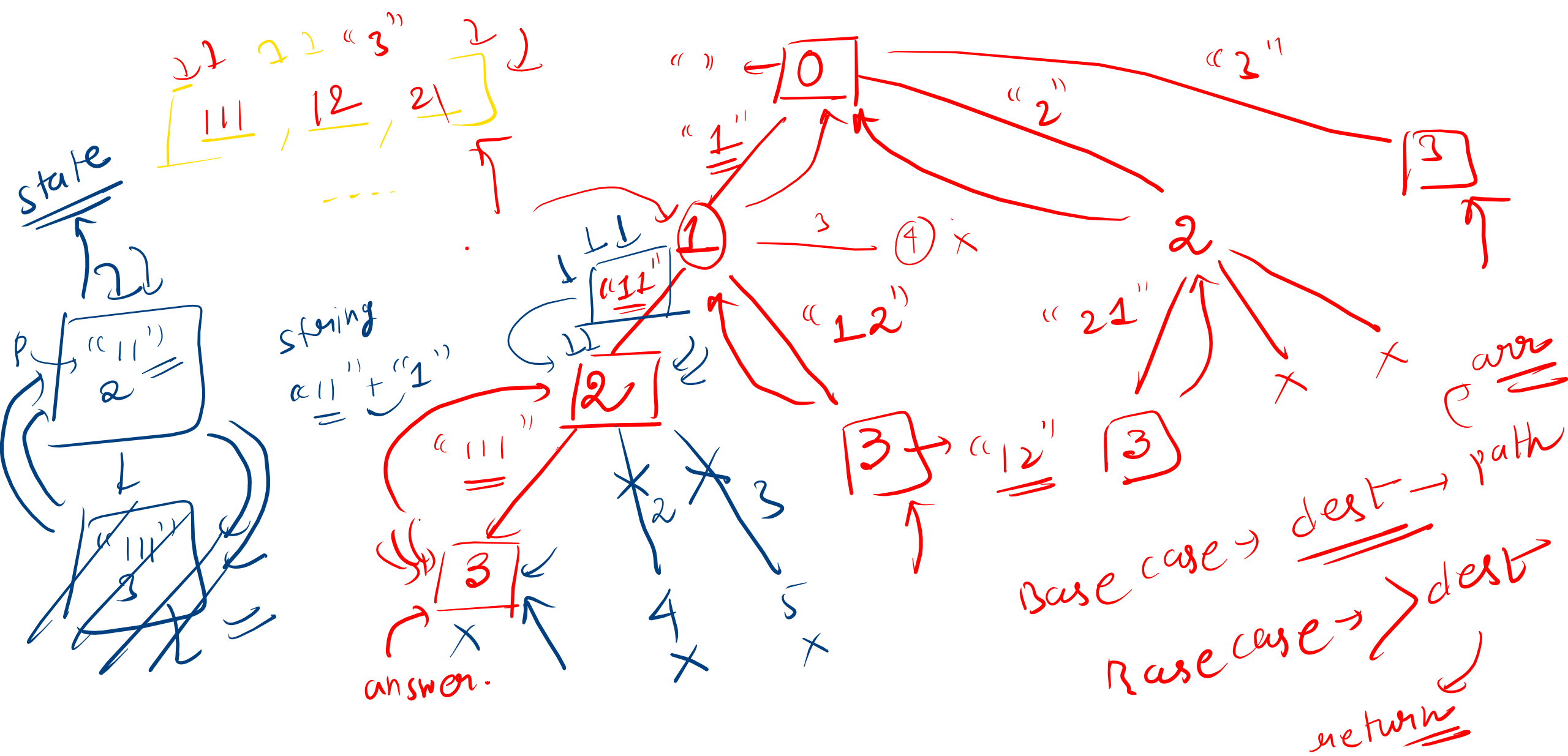


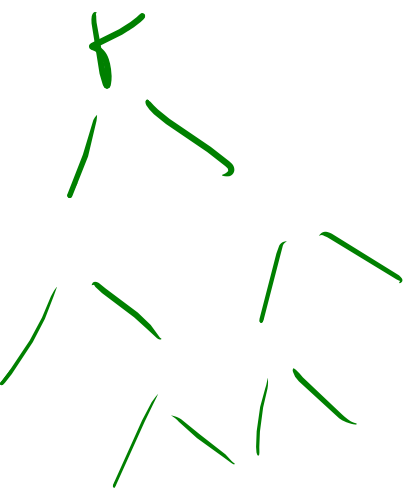
② paths



$O(2^n)$ sc $\sim \underline{\underline{n}}$







$f(s, n) \{$

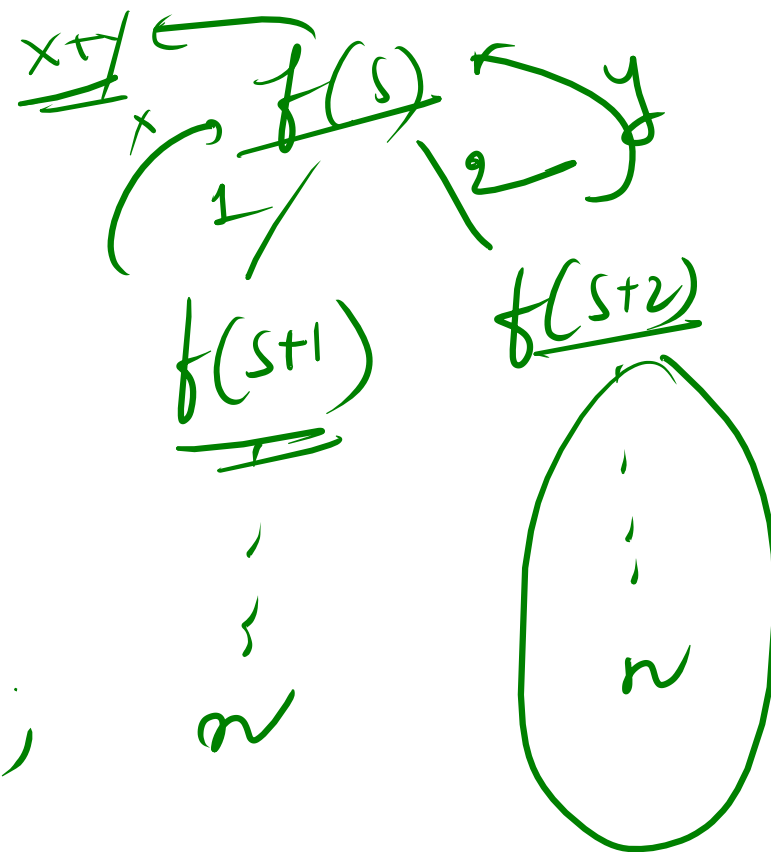
if $(s == n)$
return 1;

→ if $(s > n) \leftarrow$
return 0;

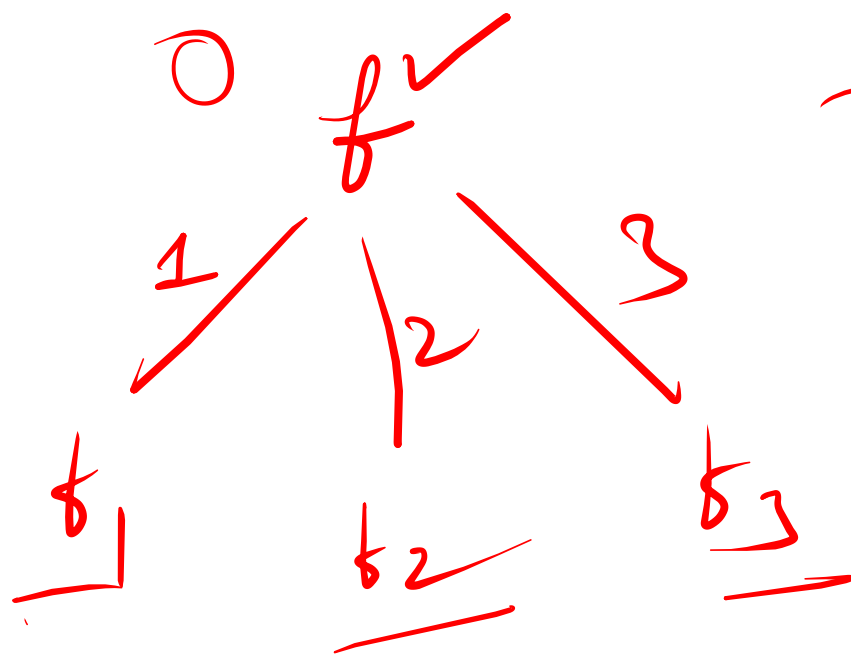
$x_1 = f(s+1, n);$

$x_2 = f(s+2, n)$

return
 $x_1 + x_2;$
}



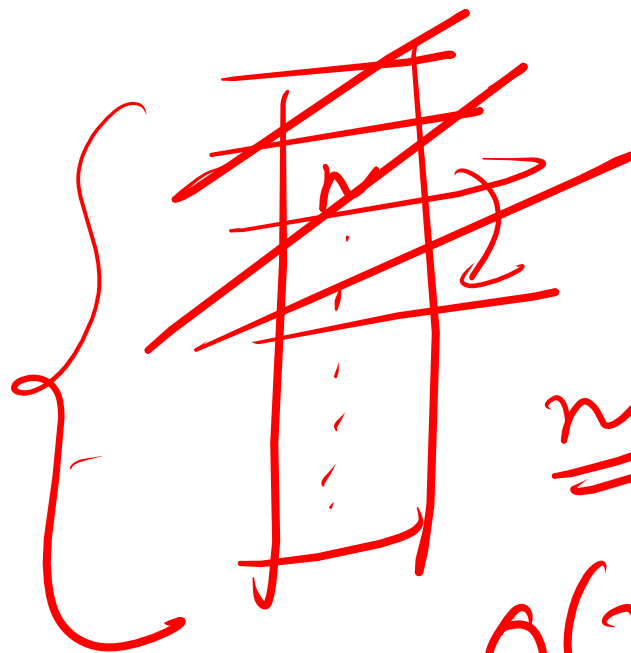
$$\underline{\underline{O(3^n)}}$$



$$\begin{aligned} &\rightarrow 3^0 + \\ &3^1 + \\ &3^2 + \\ &3^3 + 3^4 + \dots + 3^{n-1} \end{aligned}$$

$$\sim n$$

$$\underline{\underline{n}} \rightarrow \sim n + \cancel{1} - \cancel{1}$$



$$\underline{\underline{n}} \\ \underline{\underline{O(n)}}$$

$$\underline{\underline{O(3^n)}} \leq$$

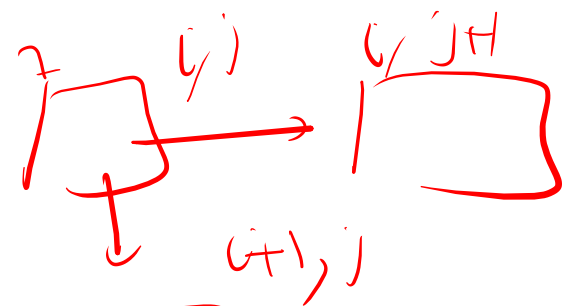
$$\frac{n=2}{m=2}$$

$$0,0 \rightarrow \underline{1,1}^2$$

how
many
paths

print

$$\frac{h}{v}^2$$



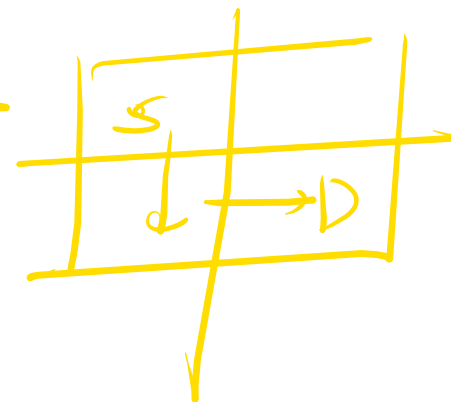
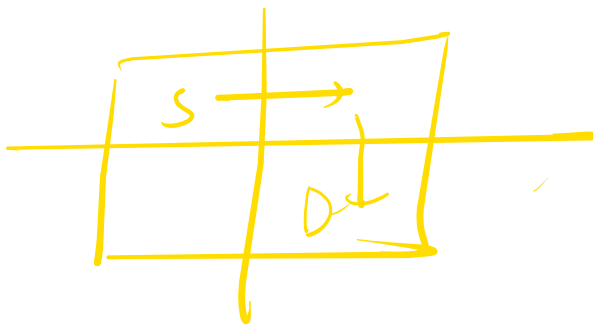
$$\underline{1,1}$$

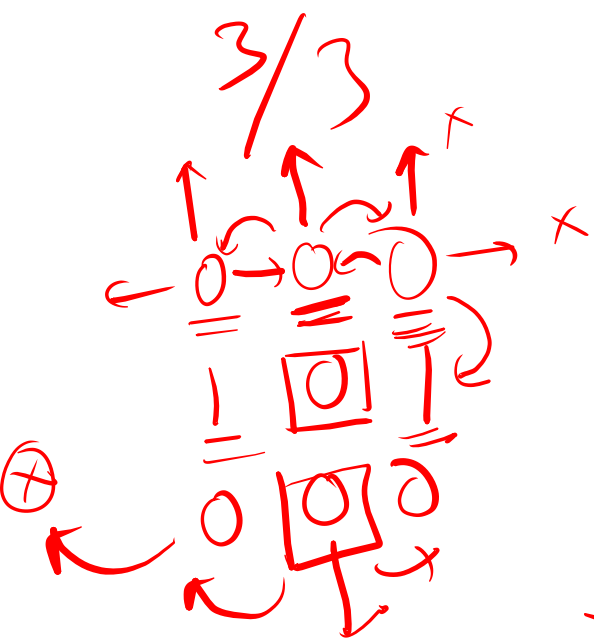
$$\underline{1,1}$$

$$\underline{2,0}$$

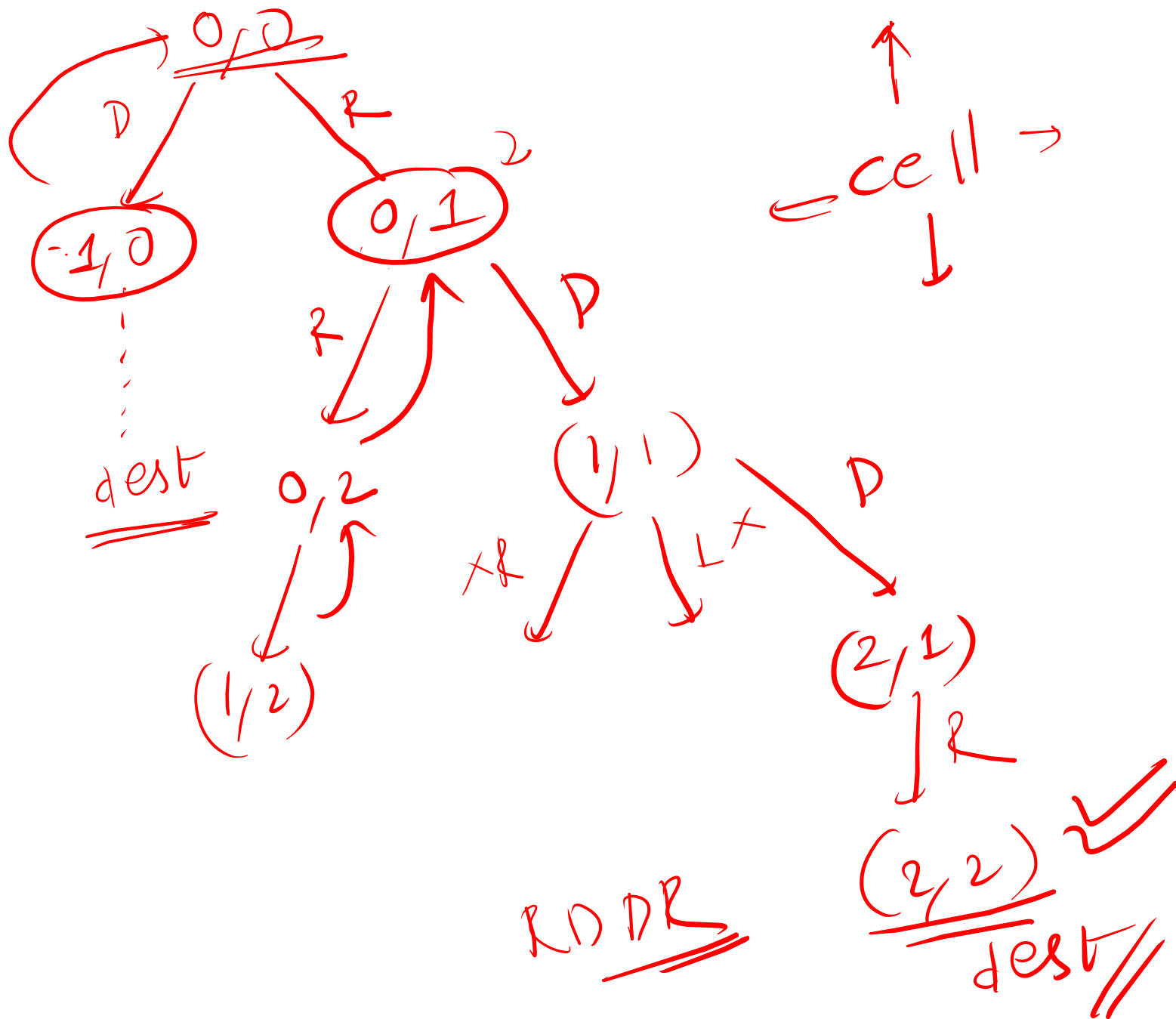
$\downarrow \rightarrow hv$

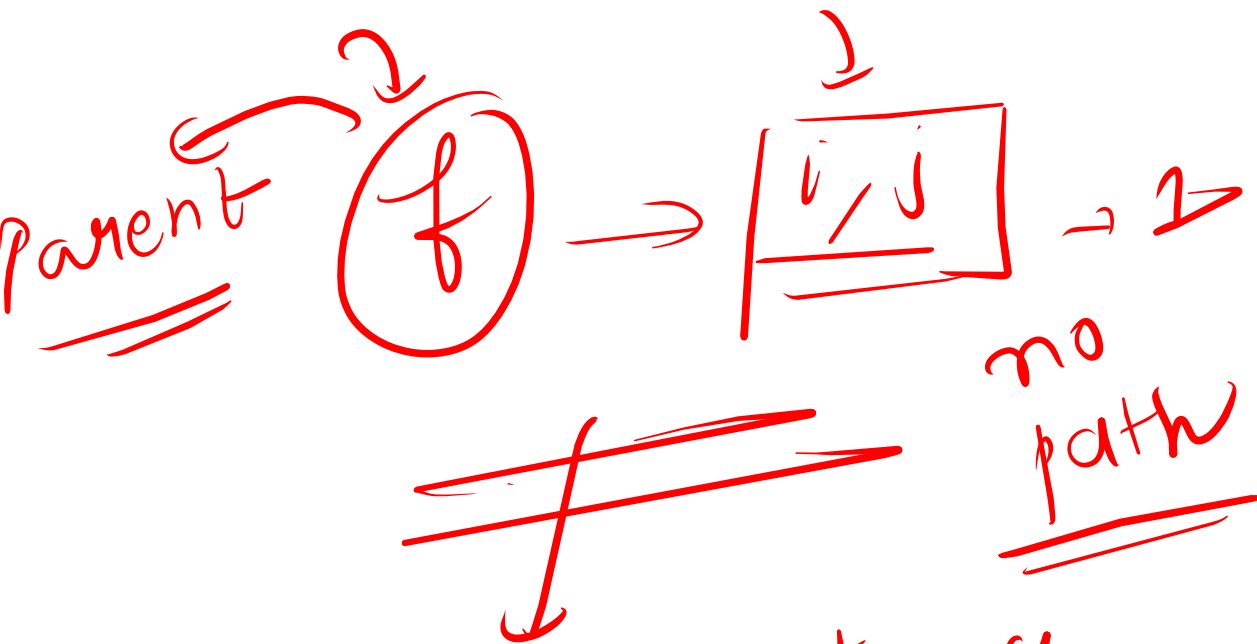
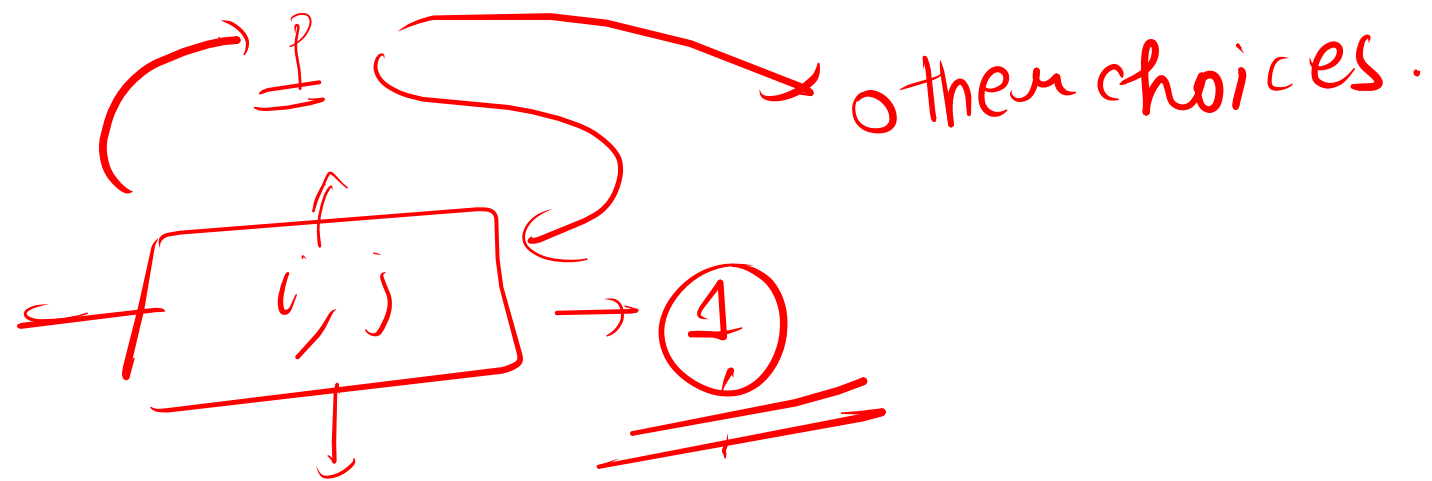
$\downarrow \leftarrow vh$



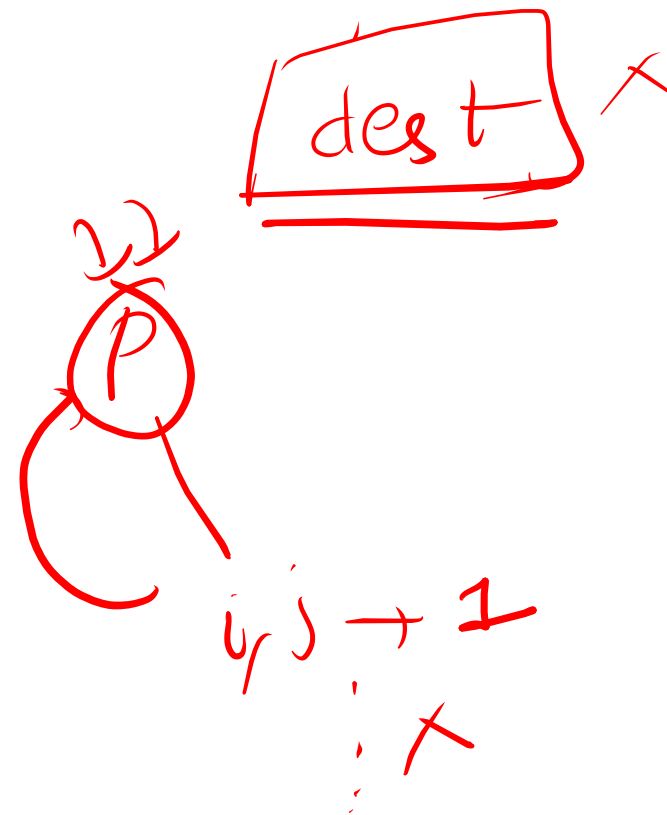


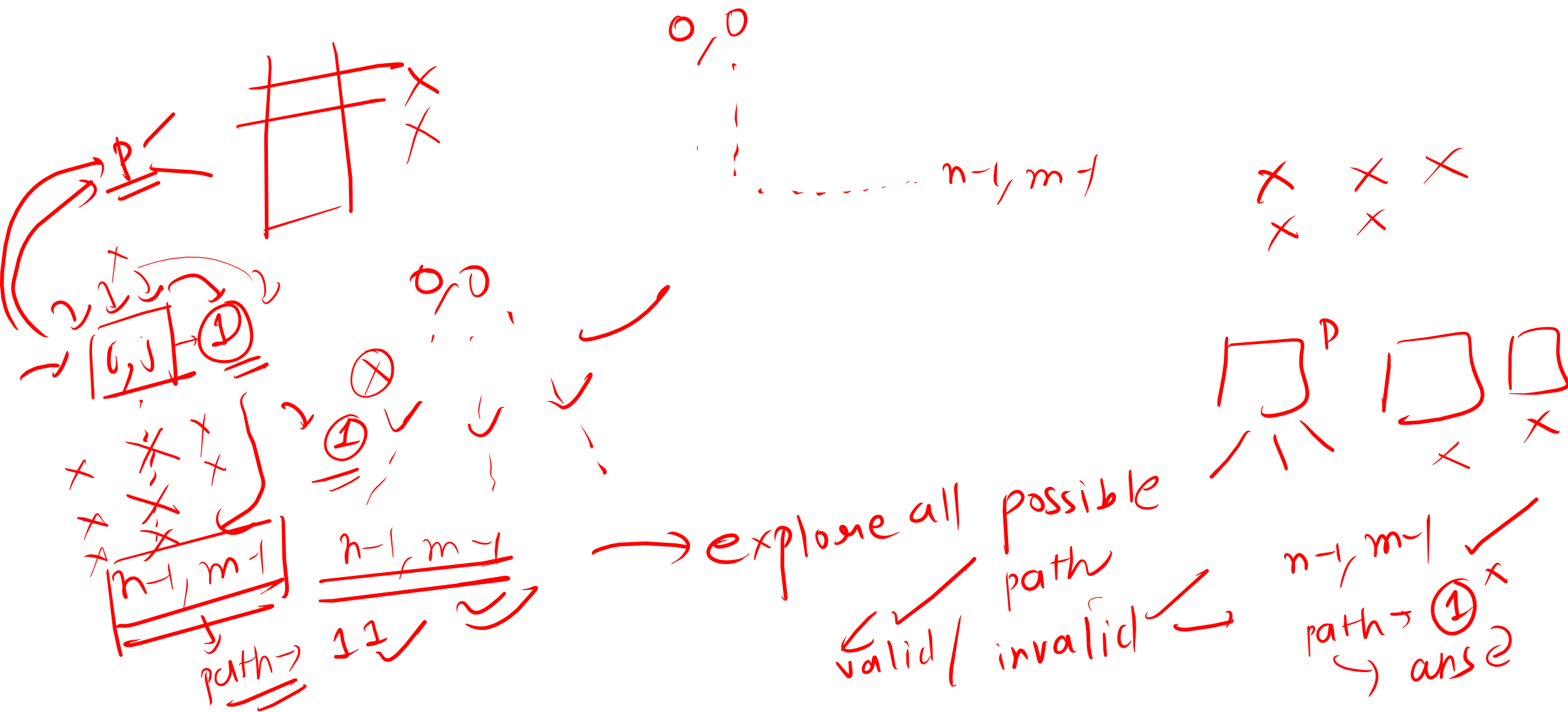
cell $\rightarrow$ visited



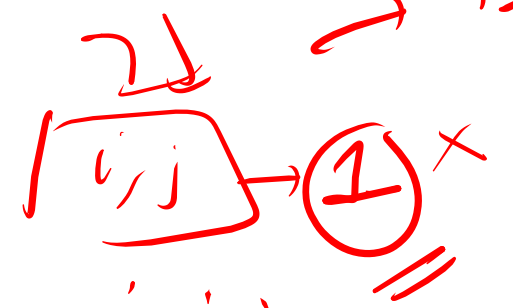


Backtracking
↳ optimising.

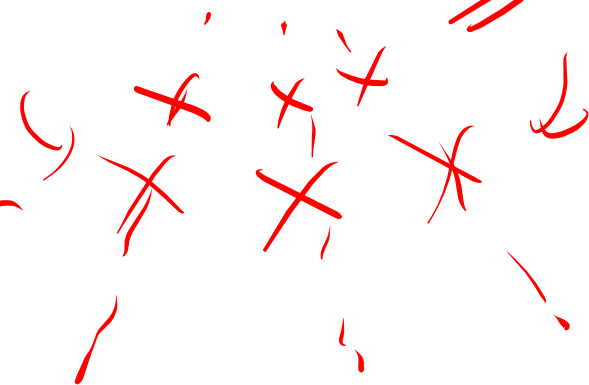




intelligent selection } → Backtracking



cut off



$n-1, m-1$

Backtracking

gen all paths

valid

~~invalid~~

ans

if/else

~~Brute force~~